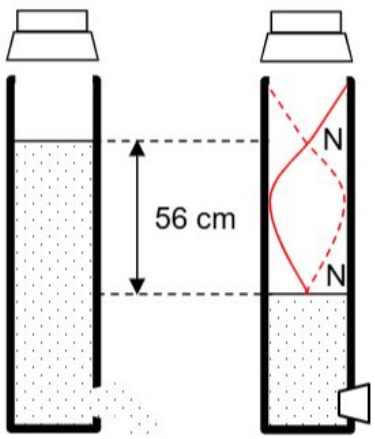
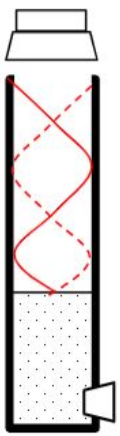
5	(a)	(i)	The dotted curve is the variation with diffraction of intensity for the single-slit diffraction pattern.	[A1]
		(ii)	$b\theta \approx \lambda$ $\lambda = (1.0\times10^{-6})(0.45)$ $= 450\times10^{-9}$ m *note, due to the gentle curve on the graph, and the usage of equation $b\sin\theta = \lambda$ which gives a slightly different result, wavelength ranges of 450 – 500 nm were accepted with 480nm being the median.	[M1] [A0]
		(iii)	Using n = 2 (second order maxima) (ecf λ from part (b)) $d\sin\theta = n\lambda$ $d = \dfrac{2(450\times10^{-9})}{\sin(0.25)}$ $= 3.6\times10^{-6}$ m Or fringe separation $x = \dfrac{3.2\times0.25}{2} = 0.4$ m $a = \dfrac{\lambda D}{x}$ $= 3.6\times10^{-6}$ m Accept a range from 3.47×10^{-6} to 3.80×10^{-6} for the value of a If n=1 is used, award 1 mark only.	[C1] [A1] [C1] [A1]
		(iv)	1. The intensity of the maxima increases or bright fringe is brighter (due to more slits). 2. Each of the maxima is narrower or fringe becomes sharper.	[B1] [B1]
	(b)	(i)	 [A1] for shape [A1] for two nodes	

		(ii)	56 cm is equivalent to half a wavelength, thus $\lambda = 0.56 \times 2 = 1.12$ m $v = f\lambda = 310 \times 1.12$ $= 347$ m s^{-1}	[C1] [A1]
		(iii)	When the frequency is increased, the next standing wave formed looks like this:  The length from the water level to the end of the tube is $56 + \frac{56}{2} = 84$ cm This length is equivalent to $\frac{5}{4}\lambda$ Thus, $\lambda = \frac{4}{5} \times 0.84 = 0.672$ m $f = \frac{v}{\lambda} = \frac{347}{0.672} = 516$ Hz	 [C1] [M1] [A1]